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Statement of Purpose

The past decade witnessed the emergence of large models. While big companies are pursuing larger and stronger of them, it is difficult for academics and individual researchers to tackle these models due to hardware insufficiency. My primary research interest is developing **efficient machine learning methods**, with the overarching aim of **making deep learning affordable and accessible** to all who utilize these technologies. To reach that goal, I am especially in favor of **efficient training/inference**, which aims to make models faster (and smaller, if possible). Particularly, I want to focus on two aspects: **system implementation** and **optimization algorithms**. These are complementary; together they construct a system in two sides: system implementation enables real-world acceleration for optimization algorithms, and optimization algorithms strike a balance between speed and accuracy.

System Implementation What are the feasible parameter types, and how to efficiently train and serve them on actual hardware?

System design influences both hardware architecture and parameter management. With NVIDIA GPUs dominating large-scale machine learning, their features (e.g., tensor cores) are well-known. My focus encompasses: 1) **Hardware and parameter aware model acceleration**: Exploring the potential of task-specific hardware (TPU, FPGA), and navigating trade-offs between speed and accuracy. For example, unstructured sparsity is difficult to accelerate, but structured sparsity may face accuracy problems. 2) **Task-specific kernel development**: Although much has been done on general kernels like GEMMs, there's still a lot to do on task specific kernels, balancing efficiency and fine-grained control, particularly in low-bit quantization and pruning. This needs to be designed along with optimization algorithms.

Through my first research project, I gained experience with CUDA kernels for efficient model pruning, and utilizing tensor cores for high-performance matrix multiplication (via Triton) has laid the groundwork for optimizing hardware and compiling high-performance kernels.

Optimization Algorithms How can we efficiently optimize a model with billions of parameters with the given structure?

Optimization algorithm needs to be designed in alignment with given structures. Typically I identify three key challenges: 1) **Poor mathematical properties**. Large models exhibit non-convexity, lacking desirable mathematical properties (e.g., small convergence radius, uncertain descent direction, large Lipschitz constant). Thus, traditional convex optimization skills will not work. 2) **Discontinuity introduced by acceleration/compression**: Continuity is the basic guarantee of gradient methods, however, models with the modified parameters (sparsity, quantization) are destined to be a discontinuous optimization target. This is because weights and activations are discrete and separate after compression. Though approximate methods, like STE, are proposed, they face oscillation problems in sparse training and quantization In this way, gradient methods also face a problem; 3) **Unpredictable acceleration/compression ratios**. For activations, observations on MoE and non-MoE models reveal that activation outliers/sparsity and weight quantization limits are data-dependent. For this reason, it is challenging to place fixed sparsity patterns on activations. For weights, the ever-increasing capacity of model knowledge also challenges the limits of quantization. For example, 4 bit weight quantization was a commonsense

in the OPT era, but on the state-of-the-art Llama3.2, even 6 bit weight seems to be losing unacceptable accuracy.

In the next five years, I want to continue pushing the boundaries of model compression on algorithms, especially optimization algorithms: 1) **Pushing the limit of low-bit quantization**: Studies on LLM ethology has demonstrated that ~2 bit is the theoretical upper bound of weight quantization (according to the quantity of knowledge it can learn). I want to move towards that ultimate goal and push the limit of low-bit weight quantization to ~2 bit. Thus, one of my subgoals is to derive new optimization methods for low-bit quantization, especially in realms of PTQ which I believe is the most promising. 2) **Making sparsity feasible and widely used**: At present the exploration of sparsity is mostly constrained to academic research because few attempts prove to be feasible and easy to use: weight sparsity brings about optimization difficulties; MoE demonstrates the huge potential of activation sparsity but may still face similar problems. In my academic career, I want to overcome optimization difficulties in weight sparsity and explore activation sparsity (e.g., MoE), KV cache, and other sparse structures. I believe with the ever-increasing size of large models, degree of sparsity increases as well, like Moore's Law.

How my past experience helps me achieve my goal in optimization algorithms for efficient training: In 2022 I joined the Tsinghua Statistical Artificial Intelligence & Learning group, supervised by Prof. Jianfei Chen and Prof. Jun Zhu. In 2023-2024 I began to explore semi-structured sparsity. I developed algorithms to accelerate pre-training via 2:4 sparsity. This was challenging because previous works are based on first-order optimization algorithms like SGD and Momentum. I devised feasible optimization methods for transformers and incorporate with popular gradient optimizers like Lion, Adam and AdamW. A series of work including two papers are published in ICML 2024 and NeurIPS 2024. **Remarkably**, my study has been cited by a study of University of Toronto & Google DeepMind as “the sparse pretraining benchmark, the state-of-the-art 2:4 pretraining method and the only semi-structured sparse pretraining work that provides end-to-end speedups.”

In 2024 summer, I joined Prof. Tuo Zhao's group in ISyE, Geogia Tech. I was taking the lead in the project, and our goal was to optimize speculative decoders on specific downstream tasks. To do this, we devised an algorithm to identify “easy” and “hard” tokens for the decoder to speculate, and proposed a workflow to differentiate them based on popular knowledge distillation methods. This experience greatly cultivated my ability to identify bottlenecks and optimize large models on a given task.

Why Berkeley I appreciate Berkeley’s academic preeminence, influential stature and reputation. Particularly, I am quite fascinated in **Prof. Kurt Keutzer**’s research on high-performance computing and machine learning, **Prof. Joseph E. Gonzalez**’s study in system support for large language model deployment, and **Prof. Ion Stoica**’s study on distributed systems and deep learning. I sincerely look forward to joining the Berkeley community, to further advance my work in Machine Learning Systems (MLSYS) and to ultimately achieve Artificial General Intelligence (AGI) for global utilization.